TEXORA · LECTURE SHEET

Yarn and Classification of Yarn

Every fancy, core-spun, textured and compact yarn in this course is a deliberate variation of an ordinary yarn. This sheet establishes what a yarn is, the characteristics that define it, and the complete classification system — by fibre, structure, twist, count and end use — on which all special yarns are built.

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 1 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. What is a yarn?

A **yarn** is a continuous strand of textile fibres, filaments, or both, assembled and held together so that it has useful tensile strength, flexibility and a reasonably uniform thickness, and can be converted into fabric by weaving, knitting, braiding or tufting. Two ideas in this definition matter. First, continuity: a yarn is effectively endless for processing purposes, supplied on a package and unwound without joins in normal use. Second, cohesion: loose fibres alone have no useful lengthwise strength — it is twist, intermingling, wrapping or adhesive bonding that locks the components together and lets the strand carry load.

In a staple-fibre yarn, strength comes mainly from **twist**. Twisting wraps fibres helically around the yarn axis; when the yarn is pulled, the helix tightens, fibres press against each other, and friction transfers load from fibre to fibre so that short fibres behave as one continuous strand. In a filament yarn the filaments are already continuous, so little or no twist is needed for strength; twist or intermingling is added mainly for cohesion, handle and appearance. Almost every special yarn in this course exploits one of these two cohesion mechanisms in a controlled, non-uniform way.

1.1 Essential characteristics of a yarn

- **Linear density (count)** — mass per unit length or length per unit mass; the primary measure of fineness (Section 5).
- **Twist level and direction** — turns per unit length and whether the helix is S or Z (Section 4).
- **Tensile strength and extension** — breaking load, usually expressed as tenacity (load per unit linear density, e.g. cN/tex), and elongation at break.
- **Evenness** — variation of mass per unit length along the yarn; measured as CV% on an evenness tester. Irregularity causes weak places and visible fabric faults in an ordinary yarn — but is created deliberately, in a controlled pattern, in slub and fancy yarns.

- **Hairiness** — fibres protruding from the yarn surface; affects pilling, dye appearance and air-jet weaving performance.
- **Appearance and handle** — lustre, bulk, softness, colour effect and surface texture, the characteristics that fancy yarns are designed to modify.

> **WHY CLASSIFICATION MATTERS**
>
> No single description covers all yarns. A buyer who orders only “20 tex yarn” may receive a single ring-spun cotton yarn, a two-ply yarn, a core-spun stretch yarn or a slub yarn — all nominally 20 tex, all behaving differently in fabric. A complete yarn specification therefore states fibre, structure, twist, count and intended use together. The five classifications below are the five axes of that specification.

2. Classification by fibre length

2.1 Staple (spun) yarns

Made from short, discontinuous fibres — cotton (typically 20–40 mm), wool, cut synthetic staple, or blends. The fibres must be opened, carded, drafted and twisted (ring, rotor, air-jet or friction spinning). Because fibre ends are everywhere in the structure, spun yarns are hairy, matte, warm and absorbent, with moderate strength limited by fibre-to-fibre friction. Faults such as thin places arise where too few fibres occupy a cross-section.

2.2 Filament yarns

Made from continuous filaments extruded through a spinneret — polyester, nylon, viscose filament, silk. A **monofilament** is a single thick filament used alone (fishing line, bristles, some technical fabrics). A **multifilament** yarn bundles many fine filaments (e.g. 167 dtex / 48 filaments): smooth, lustrous, strong per unit mass and low in hairiness, but flat and cold in handle unless textured (Topic 6). Filament yarns may be flat (untwisted), twisted for cohesion, or intermingled by air jets.

FEATURE	STAPLE (SPUN) YARN	FILAMENT YARN
Fibre length	Short, discontinuous (mm range)	Continuous (km range)
Cohesion	Twist essential for strength	Filaments continuous; twist optional
Surface	Hairy, matte	Smooth, lustrous (flat yarn)
Strength mechanism	Fibre friction under twist	Full filament strength
Typical products	Cotton shirting, denim, knitwear	Lining, hosiery, sportswear, sewing thread core

Many special yarns are **hybrids**: a filament core wrapped in staple sheath (core-spun yarn, Topic 5), a filament wrapped by another yarn (wrap yarn, Topic 4), or staple and filament combined in a fancy effect. Classification by fibre length therefore describes each component, not always the whole yarn.

3. Classification by structure

3.1 Single yarn

One strand as it leaves the spinning or extrusion process: one spun strand, or one bundle of filaments. It is the building block for all higher structures. A single yarn normally carries twist in one direction and is therefore torque-unbalanced — a cut length held slack will snarl — unless it has been set.

3.2 Ply or folded yarn

Two or more single yarns twisted together, written e.g. 20 tex × 2. Plying usually uses twist opposite in direction to the singles twist, which balances torque, rounds the cross-section, and improves evenness and abrasion resistance because a weak place in one single is bridged by the other. Sewing threads, knitting yarns and quality warp yarns are commonly two-ply or three-ply.

3.3 Cord or cabled yarn

Two or more ply yarns twisted together — twisting a twisted structure a second time. Cords are strong, round and abrasion-resistant: tyre cord, rope and industrial sewing thread. Each successive twisting stage alternates direction in a balanced cord so the final structure does not kink.

3.4 Novelty (fancy / effect) yarns

Yarns in which irregularity of thickness, colour, loop, knot or material is introduced deliberately and repeatably for decorative effect: loop, slub, snarl, spiral, chenille, gimp and many others. They are the subject of Topics 2 and 3. Structurally most consist of an effect component, a core component and often a binder — a fundamentally different architecture from a uniform single yarn.

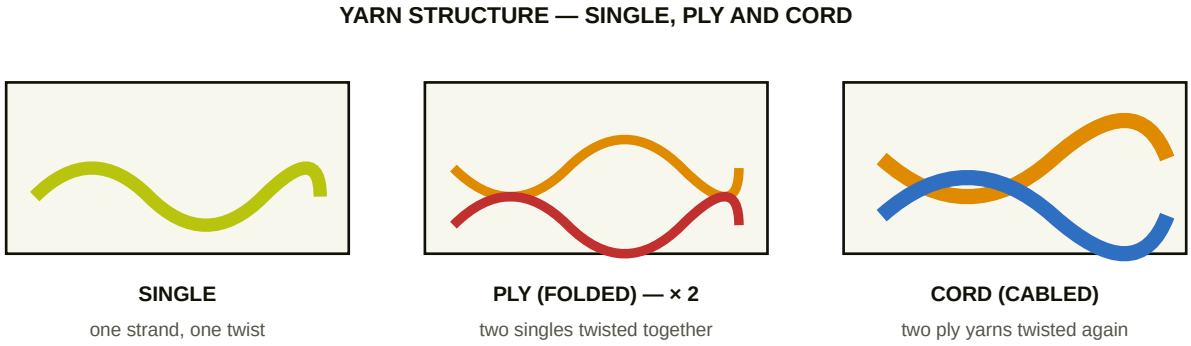


Figure 1 — The three levels of yarn structure. A cord is a ply of plies; a novelty yarn departs from all three by adding a deliberate decorative irregularity.

4. Classification by twist

4.1 Twist direction — S and Z

Hold a yarn vertically. If the visible helices slope in the same direction as the central stroke of the letter **Z** (rising to the right, \), the yarn has **Z twist**; if they slope like the central stroke of **S** (rising to the left, /), it has **S twist**. Most spinning produces Z-twist singles; plying is then done in S direction. Direction matters in fabric: mixing S- and Z-twist yarns changes light reflection and is used deliberately in crepe and shadow effects, and a wrong-direction component in a fancy yarn will untwist instead of locking.

4.2 Twist amount

Twist is expressed as turns per unit length (turns/m or turns/inch, tpi) or through the twist multiplier, which links twist to count so yarns of different fineness can be compared:

$$\text{Twist (turns/m)} = \text{twist multiplier } \alpha_{tex} \times 1000 / \sqrt{(\text{tex})} \quad \cdot \quad \text{in cotton count: tpi} = \text{twist factor} \times \sqrt{(\text{Ne})}$$

By amount, yarns fall into broad families: **low-twist / soft-twist** yarns (bulky, soft, weaker — knitting yarns, filling), **medium-twist** yarns (general-purpose weaving and knitting), **hard-twist** yarns (lean, strong, lively) and **crepe-twist** yarns (very high twist, deliberately torque-lively so the fabric puckers into a crepe surface after finishing). Twist amount is therefore both a strength variable and an appearance variable.

5. Classification by count (fineness) and count systems

Yarns are too fine to specify by diameter reliably — diameter changes with twist and compression — so fineness is specified by linear density. Two families of systems exist, and confusing them reverses

the meaning of the number.

5.1 Direct systems — higher number = coarser

- **Tex** — mass in grams of 1,000 m. The universal SI-based system. Sub-units: decitex (dtex, per 10,000 m, used for filaments) and millitex; multiples: kilotex for sliver/roving-scale strands.
- **Denier** — mass in grams of 9,000 m, traditional for filament yarns. Denier = 9 × tex.

5.2 Indirect systems — higher number = finer

- **English cotton count, Ne** — number of 840-yard hanks per pound. Still dominant in the cotton and denim trade.
- **Metric count, Nm** — metres per gram × 1,000 formulation: number of 1,000 m lengths per kilogram (equivalently, metres per gram × 1000 /1000 — in practice Nm = length in m per g weight numerically equal to km per kg). Nm = 1,000 / tex.
- Other regional indirect systems (worsted count, linen lea) survive in their own trades but convert through the same principle: length per unit mass.

SYSTEM	TYPE	DEFINITION	CONVERSION TO TEX
Tex	Direct	g per 1,000 m	—
Decitex (dtex)	Direct	g per 10,000 m	tex = dtex / 10
Denier	Direct	g per 9,000 m	tex = denier / 9
Ne (cotton)	Indirect	840-yd hanks per lb	tex = 590.5 / Ne
Nm (metric)	Indirect	km per kg	tex = 1,000 / Nm

WORKED EXAMPLE — CONVERTING 20 TEX

A yarn is specified as 20 tex. Convert to Ne, Nm and denier.

Ne = 590.5 / tex = 590.5 / 20 = **29.5 Ne** (29.53 before rounding).

Nm = 1,000 / tex = 1,000 / 20 = **50 Nm**.

Denier = 9 × tex = 9 × 20 = **180 denier**; in decitex, 20 tex = **200 dtex**.

Check the logic: tex is direct, so a larger tex would give a *smaller* Ne and Nm — the indirect numbers fall as the yarn gets coarser. All four numbers describe exactly the same yarn.

By fineness, trade language loosely groups yarns as **coarse** (e.g. below about 10 Ne), **medium** (roughly 10–40 Ne) and **fine** (above about 40 Ne), but the boundaries differ by fibre and product, so the number — never the adjective alone — is the specification. For folded yarns the resultant count must be stated: two 20 tex singles plied with negligible contraction give approximately 40 tex resultant, written 20 tex

× 2; in indirect notation the same yarn is written Ne 29.5/2, and its resultant Ne is approximately half the singles Ne.

6. Classification by end use

- **Weaving yarns** — warp yarns need higher strength, abrasion resistance and low hairiness for loom performance, and are often sized; weft (filling) yarns can be softer and bulkier.
- **Knitting yarns** — softer twist, good evenness and low friction for loop formation; hosiery, jersey and sweater yarns.
- **Sewing threads** — a product class of their own: ply or core-spun structures, high tenacity, controlled elongation and excellent abrasion resistance.
- **Hand-knitting and craft yarns** — sold by appearance and handle in small packages; fancy structures are common.
- **Industrial / technical yarns** — tyre cord, rope, conveyor-belt, filtration, geotextile and composite-reinforcement yarns, specified by performance rather than appearance.
- **Carpet and furnishing yarns** — bulky, resilient, often textured or fancy, chosen for surface effect and wear.

End use closes the specification loop: the same base yarn finished for warp use (sized, hard twist, low hairiness) and for knitting use (waxed, soft twist) is, commercially and technically, two different yarns. Special yarns are almost always designed backwards from an end use — stretch denim demands core-spun yarn, a boucle jacket fabric demands a loop fancy yarn — which is why this course proceeds from classification to construction.

Glossary

Yarn — continuous cohesive strand of fibres and/or filaments suitable for fabric formation. **Staple yarn** — yarn spun from short discontinuous fibres. **Filament yarn** — yarn of continuous filaments. **Single yarn** — one strand as spun or extruded. **Ply (folded) yarn** — two or more singles twisted together. **Cord (cabled) yarn** — two or more ply yarns twisted together. **S twist / Z twist** — helix direction matching the central stroke of S or Z. **Tex** — grams per 1,000 m (direct). **Ne** — 840-yard hanks per pound (indirect). **Tenacity** — breaking load per unit linear density. **Hairiness** — fibres protruding from the yarn surface. **Novelty yarn** — yarn with deliberate decorative irregularity.

Key points

• A yarn gains useful strength from cohesion — twist friction in staple yarns, continuity in filament yarns.

• Classification runs on five axes: fibre length, structure, twist, count and end use — a full specification needs all five.
• Structure builds up single → ply → cord, each stage normally twisted opposite to the last for balance.
• Direct systems (tex, denier): higher number = coarser. Indirect systems (Ne, Nm): higher number = finer.
• Conversions: tex = 590.5 / Ne = 1,000 / Nm; denier = 9 × tex; 20 tex = 29.5 Ne = 50 Nm = 180 denier.
• Novelty yarns are a structural class of their own and form the core subject of this course.

Practice

1. Convert 15 tex to Ne, Nm and denier, showing each formula.
2. A sewing thread is made by plying three 12 tex singles, then cabling two of those ply yarns. State the structure name at each stage and the approximate resultant tex of the final cord, ignoring contraction.
3. Explain why two yarns both labelled “30 Ne” can still be unsuitable substitutes for each other. Name at least four specification details beyond count that must match.
4. A vertical yarn shows helices rising to the left, like the centre stroke of the letter S. State its twist direction, and state the usual plying direction used to balance such a single if it had been spun in the more common direction instead.

Answer key

1. Ne = 590.5 / 15 = **39.4 Ne**. Nm = 1,000 / 15 = **66.7 Nm**. Denier = 9 × 15 = **135 denier** (= 150 dtex).
2. Stage 1: three singles twisted together = a **ply (folded) yarn**, approx. 3 × 12 = **36 tex**. Stage 2: two ply yarns twisted together = a **cord (cabled) yarn**, approx. 2 × 36 = **72 tex** resultant.
3. Count fixes only linear density. Fibre type and staple length, single vs. ply structure, twist direction and amount, evenness, hairiness, strength/tenacity and finish (sized, waxed, dyed) can all differ — any one of them can change loom performance, fabric appearance or handle, so the full five-axis specification must match, not the count alone.
4. The yarn has **S twist**. The more common singles direction is Z, and such a single is normally plied in **S** direction — plying opposite to the singles twist — to balance torque and round the ply yarn.
